Master Thesis supplement: Joint Modelling

"Towards Bayesian Joint Modelling of Ranks and Scores in the Assessment of Grant Proposal Quality"

Zürich, November 6, 2026

Introduction

This document describes a novel Bayesian approach to jointly model scoring and ranking data using the `inlajoint` package (Rustand et al. 2024). While a frequentist approach to joint modelling already exists in the form of the Mallows-Binomial model (Pearce and Erosheva 2022), a Bayesian model has yet to be proposed in this context. We define and describe this novel Bayesian model. Then, we apply it on the Gallo et al. (2023) dataset and report the results it provides. Finally, we discuss the model's limitations and the further work necessary to validate it for use in real-world scenarios. During the model fitting, we provide all code necessary to reproduce the analyses performed in this document.

Model Definition

To unify the scale of the scoring and ranking data, we first transform the data through z-scoring as follows:

$$y^z_{i,j} = \frac{y_{ij} - \hat{\bar{y}}}{\hat{\sigma}_y}$$

$$r^z_{i,j} = \frac{r_{ij} - \hat{\bar{r}}}{\hat{\sigma}_r}$$

Using the standardised $y^z_{i,j}$ data, we modify the Bayesian hierarchical model for scores defined by Heyard et al. (2022) in order to obtain:

$$y^z_{i,j} = \beta^y_0 + \theta^y_i + \lambda^y_j$$

$$\theta^y \sim N(0, \sigma^2_{\theta^y}) \qquad \beta^y_0 \sim N(\mu^y, \sigma^2_y)$$

where θ^y_i represents the latent qualities of an object o_i according to the transformed scoring data, λ^y_j represents the assessor-specific differences in strictness and scoring habit, and β^y_0 represents and estimates the mean standardised score of all objects.

Analogously, we create a second model for the transformed ranking data as follows:

$$y^z_{i,j} = \beta^y_0 + \theta^r_i$$
$$\theta^r \sim N(0, \sigma^2_{\theta^r}) \qquad \beta^r_0 \sim N(\mu^r, \sigma^2_r)$$

where θ^r_i represents the latent qualities of an object o_i according to the transformed ranking data, and β^r_0 represents and estimates the mean standardised rank of all objects.

We join these models by defining a correlation constraint ($\text{corr}(\theta^y, \theta^r) = \rho_\theta$) between the random effects of both models, which are effectively the latent quality variables. The `inlajoint` package does this automatically, and fits both models simultaneously taking this correlation into account. This provides the final model:

$$\begin{cases} y^z_{i,j} = \beta^y_0 + \theta^y_i + \lambda^y_j \\ r^z_{i,j} = \beta^r_0 + \theta^r_i \end{cases}$$
$$\theta^y \sim N(0, \sigma^2_{\theta^y}) \qquad \theta^r \sim N(0, \sigma^2_{\theta^r})$$
$$\text{corr}(\theta^y, \theta^r) = \rho_\theta$$
$$\beta^y_0 \sim N(\mu^y, \sigma^2_y) \qquad \beta^y_0 \sim N(\mu^r, \sigma^2_r)$$

Model Fitting & Results

In the Gallo et al. dataset, there are 259 scoring observations, but only 49 ranking observations. This provides us with three paths for model fitting: (1) We analyse the data as-is and ignore the missingness of the ranking data. (2) We analyse only the complete cases in the data and remove 210 scoring observations. (3) We correct for the difference in the amount of observations in the model. While path (3) is the most promising for real-world applications, it is beyond the scope of this thesis. We therefore only provide examples of models fitted using path (1) and path (2).

As-is Analysis

The code below was used to fit the joint model:

```
# Overview of data
kable(head(dat_as_is))
```

RandomRevID	grant_id	std_score	std_rank
225	1	0.023	NA
218	1	0.023	NA
145	1	NA	NA
235	1	0.023	NA
308	1	0.023	NA
203	1	0.023	NA

```
# Fitting model
inla_sr_joint_as_is <- joint(
  formLong = list(
    std_score  ~ 1 + (1 | grant_id) + (1| RandomRevID),
    std_rank  ~ 1 + (1 | grant_id)
  ),
  dataLong = dat_as_is, id = "grant_id",
  family  = c("gaussian", "gaussian"),
  corLong = T
)
```

The resulting posterior mean estimates of the 10 best objects' latent qualities in the as-is model are presented in the table below:

	x17	x25	x19	x21	x4	x23	x22	x20	x6	x13
Estimate	-1.28	-1.22	-1.22	-1.13	-0.96	-0.90	-0.86	-0.85	-0.76	-0.72

Complete Cases

The code below was used to fit the joint model:

```
# Overview of data
kable(head(dat_c_cases))
```

	RandomRevID	grant_id	std_score	std_rank
27	269	3	-0.638	1.432
31	225	4	-0.638	-0.293
32	218	4	0.663	0.857
36	203	4	1.313	0.857
37	269	4	1.313	0.857
38	321	4	-0.638	0.857

```
# Fitting model
inla_sr_joint_complete <- joint(
  formLong = list(
    std_score  ~ 1 + (1 | grant_id) + (1| RandomRevID),
    std_rank  ~ 1 + (1 | grant_id)
  ),
  dataLong = dat_c_cases, id = "grant_id",
  family  = c("gaussian", "gaussian"),
  corLong = T
)
```

The resulting posterior mean estimates of the 10 best objects' latent qualities in the complete cases model are presented in the table below:

	x17	x25	x19	x21	x3	x4	x23	x13	x22	x16
Estimate	-1.18	-0.93	-0.91	-0.59	-0.50	0.34	0.47	0.59	0.59	0.91

Discussion and Limitations

In this supplement, we present how the `inlajoint` package can be used in combination with scoring and ranking data to create a joint model assessing grant proposal qualities. While this model is still in its early stages, we provide all code snippets necessary to use, reproduce, and extend it.

Much work still remains to be done before this joint model can be used in practice. First of all, in our model definition, we transform the ranking data using z-scoring. Performing such transformations on ranking data is a polarising topic in the literature (Stevens 1946; Mosteller 1977; Velleman and Wilkinson 1993). Whether this can be avoided, and what other possible model formulations exist still needs to be clarified. Then, due to the nature of grant proposal review datasets, it is very likely that real world datasets will have many missing data points. Giving rankings to objects is a cumbersome task, and in consequence only partial ranking data will be available in most real-world scenarios. Strategies to deal with incomplete data also need to be investigated. Finally, the `inlajoint` package is very flexible, and many options exist to define the correlation structure of the model. These options should also be explored when creating a joint model.

Future work on this model could compare the Bayesian joint approach presented in this supplement to other already existing models used to analyse scoring and ranking data. Moreover, analysis on simulation data could be used to validate and quantify the performance of this model. While the work presented in this supplement is only preliminary, it can serve as a stepping stone for further work in implementing such joint models in the assessment of grant proposal qualities.

Bibliography

Gallo, Stephen A., Michael Pearce, Carole J. Lee, and Elena A. Erosheva. 2023. "A New Approach to Grant Review Assessments: Score, Then Rank." *Research Integrity and Peer Review* 8 (1): 10. https://doi.org/10.1186/s41073-023-00131-7.

Heyard, Rachel, Manuela Ott, Georgia Salanti, and Matthias Egger. 2022. "Rethinking the Funding Line at the Swiss National Science Foundation: Bayesian Ranking and Lottery." *Statistics and Public Policy* 9 (1): 110–21. https://doi.org/10.1080/2330443X.2022.2086190.

Mosteller, Frederick. 1977. *Data Analysis and Regression : A Second Course in Statistics*. Reading, Mass. : Addison-Wesley Pub. Co. http://archive.org/details/dataanalysisregr0000most.

Pearce, Michael, and Elena A. Erosheva. 2022. "A Unified Statistical Learning Model for Rankings and Scores with Application to Grant Panel Review." *Journal of Machine Learning Research* 23 (210): 1–33. http://jmlr.org/papers/v23/21-1262.html.

Rustand, Denis, Janet van Niekerk, Elias Teixeira Krainski, Håvard Rue, and Cécile Proust-Lima. 2024. "Fast and Flexible Inference for Joint Models of Multivariate Longitudinal and Survival Data Using Integrated Nested Laplace Approximations." *Biostatistics* 25 (2): 429–48. https://doi.org/10.1093/biostatistics/kxad019.

Stevens, S. S. 1946. "On the Theory of Scales of Measurement." *Science* 103 (2684): 677–80. https://doi.org/10.1126/science.103.2684.677.

Velleman, Paul F., and Leland Wilkinson. 1993. "Nominal, Ordinal, Interval, and Ratio Typologies Are Misleading." *The American Statistician* 47 (1): 65–72. https://doi.org/10.1080/00031305.1993.10475938.